

**Definition 9.1**

A group  $G$  is cyclic if there is an element  $a \in G$  such that

$$G = \{a^n \mid n \in \mathbb{Z}\}$$

or, in other notation,  $G = \langle a \rangle$ . In such case we say that  $a$  is a *generator* of  $G$ .

**Theorem 9.2**

If  $G$  is a finite group then  $G$  is cyclic if and only if there is an element  $a \in G$  such that  $|a| = |G|$ .

**Theorem 9.3**

Every subgroup of a cyclic group is cyclic.

#### Theorem 9.4

If  $G$  is a finite cyclic group and  $H \subseteq G$  is a subgroup then  $|H|$  divides  $|G|$ .

#### Theorem 9.5

If  $G$  is a finite cyclic group and  $d > 0$  is an integer that divides  $|G|$  then there exists exactly one subgroup  $H \subseteq G$  such that  $|H| = d$ .

### Theorem 9.6

Let  $G = \langle a \rangle$  be a cyclic group of order  $n$ . An element  $a^k$  is a generator of  $G$  (i.e.  $\langle a^k \rangle = G$ ) if and only if  $\gcd(n, k) = 1$ .

**Exercise 9.7.** In the group  $\mathbb{Z}_{15}$  find all elements  $a$  such that  $a$  generates  $\mathbb{Z}_{15}$

### Theorem 9.8

Let  $G_1 = \langle a_1 \rangle$  and  $G_2 = \langle a_2 \rangle$  be finite cyclic groups. The group  $G_1 \times G_2$  is cyclic if and only if  $\gcd(|G_1|, |G_2|) = 1$ .

### Theorem 9.9

For  $i = 1, \dots, n$  let  $G_i = \langle a_i \rangle$  be a cyclic group. The group  $G_1 \times G_2 \times \dots \times G_n$  is cyclic if and only if  $\gcd(|G_i|, |G_j|) = 1$  for all  $i \neq j$ .